

Active Management of IO Beta

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IO Beta Allocation Strategy: July 2016

Commentary

- ▶ **IO Index Returns:** 4.15% for Index with Curve (EDFs+Tsy Futures) and TBA Hedges (July 2016)
- ▶ **Fund Beta Allocation YTD:** Underperformed IO Index (w/ Curve Hedges) by 642bps

IO Beta Allocation Strategy: August 2016 Outlook

- ▶ **Fixed Income Fund Allocation:** 35% (July: 35%)
- ▶ **Model 1 (IO Px/Cpn not duration-adjusted):** 64%
(June: 36%)
- ▶ **Model 2 (IO Px/Cpn duration-adjusted):** 41%
(June: 57%)

Active Management of IO Beta

Investigate if we can design an **active** strategy for investing in IO Beta that outperforms a **passive** strategy that is always 100% invested.

Metrics and Set-up

- ▶ Create a **Hedged IO** index
 - ▶ Hedge various IOS with current coupon TBA using model durations (Barclays) until December 2014. From 2015 onwards, use YB durations and employ TBA and Curve hedges.
 - ▶ Index is created by using monthly market value weights for each IOS based on the market weight of the corresponding cohort
 - ▶ Ginnie Mae IOS are excluded
- ▶ Simulating Passive IO Beta: Total Return and Sharpe Ratio of Hedged IO index from March 2011 - July 2016
- ▶ Simulating Active IO Beta: Dynamic allocation to Hedged IO index based on relative value metrics

Performance of Passive IO Beta Strategy: Mar 2011 - July 2016

- ▶ **CAGR:** 8.4%
- ▶ **Ann. Volatility:** 14.7%
- ▶ **Sharpe Ratio:** 0.6
- ▶ **% Positive Months:** 60
- ▶ **Best Month:** 11.6%
- ▶ **Worst Month:** -12.9%
- ▶ **Max Drawdown:** -23.5%
- ▶ **Drawdown Period:** June 2011 - February 2013

Framework for Active IO Beta Strategy

- ▶ Regress Hedged IO Index returns against RV metrics (1m lag):
 - ▶ IO Index Price/Coupon¹
 - ▶ Projected 1-month Carry²
 - ▶ Convexity: 60-day moving average of realized volatility on the current coupon
 - ▶ Prepayment Risk: Incentive for the Mortgage Universe
 - ▶ Yield Curve Slope: 10s-2s slope
- ▶ Calculate z-score of projected return and allocate to Beta (Net IO) based on score

¹While OAS would also be useful, in practice dealer model updates are not always appropriately back-filled which makes it challenging to get a consistent time series of values.

²This is not available on a historical basis so we use realized carry as a proxy.

Active IO Beta Strategy: Regression Results

- ▶ Significant Variables
 - ▶ High: IO Index Price/Coupon, Prepayment Risk
 - ▶ Medium: Realized Volatility
 - ▶ Low: Realized Carry, Yield Curve Slope, Index OAS
- ▶ Final Model
 - ▶ Proj. Return $\sim P_x/Cpn + Cnvx + PrepayRisk + Carry$

Active IO Beta Strategy: Beta Allocation

- ▶ Solve for $k = a + bz$ ($z = z\text{-score}$) such that we optimize the Sharpe Ratio of the strategy $k \times \text{Hedged IO Return}$.
- ▶ Set $\text{IO Beta}_{\max} = 100\%$ and $\text{IO Beta}_{\min} = -0.25\%$
- ▶ Rescale k so that it is (more or less) mapped to these boundaries.

Active IO Beta Strategy: Two Beta Allocation Models

- ▶ Model 1: Don't adjust IO Index P_X/C_{pn} for Interest Rates. $R^2 = 0.33$.
- ▶ Model 2: Adjust IO Index P_X/C_{pn} for Interest Rates.³ $R^2 = 0.18$.

³We estimate that the P_X/C_{pn} ratio moves by about 0.14 for each 10bps move in the current coupon.

Active IO Beta Strategy: Model 1 Return Profile

- ▶ **CAGR:** 9.0%
- ▶ **Ann. Volatility:** 5.3%
- ▶ **Sharpe Ratio:** 1.7
- ▶ **% Positive Months:** 77
- ▶ **Best Month:** 9.6%
- ▶ **Worst Month:** -0.8%
- ▶ **Max Drawdown:** -0.9%
- ▶ **Drawdown Period:** November 2011 - January 2012

Active IO Beta Strategy: Model 2 Return Profile

- ▶ **CAGR:** 6.8%
- ▶ **Ann. Volatility:** 5.3%
- ▶ **Sharpe Ratio:** 1.3
- ▶ **% Positive Months:** 73
- ▶ **Best Month:** 9.6%
- ▶ **Worst Month:** -3.0%
- ▶ **Max Drawdown:** -3.6%
- ▶ **Drawdown Period:** September 2011 - January 2012

Track Record of IO Beta Allocation Strategies

	IO Beta Allocation			100% Beta CrvAdj	Fund	Model1	Model2	100% Beta CrvAdj	Fund	Model1	Model2
	Fund	Model1	Model2								
Jan-15	26%	-11%	-16%	-8.9%	-2.31%	0.98%	1.40%				
Feb-15	25%	26%	59%	5.5%	1.38%	1.44%	3.24%				
Mar-15	21%	-7%	-51%	1.2%	0.25%	-0.08%	-0.61%				
Apr-15	15%	-21%	-16%	1.2%	0.18%	-0.26%	-0.19%				
May-15	22%	-28%	-36%	-1.7%	-0.37%	0.47%	0.61%				
Jun-15	31%	-17%	-24%	-0.7%	-0.20%	0.11%	0.16%				
Jul-15	33%	-25%	-51%	1.2%	0.40%	-0.30%	-0.61%				
Aug-15	28%	-42%	-16%	-1.5%	-0.42%	0.64%	0.23%				
Sep-15	24%	-24%	-23%	-1.6%	-0.38%	0.39%	0.37%				
Oct-15	26%	-10%	5%	-0.1%	-0.03%	0.01%	-0.01%				
Nov-15	21%	2%	-5%	1.8%	0.38%	0.04%	-0.08%				
Dec-15	22%	15%	0%	1.4%	0.30%	0.21%	0.01%	-2.23%	-0.85%	3.66%	4.53%
Jan-16	17%	12%	3%	-2.1%	-0.36%	-0.25%	-0.07%				
Feb-16	31%	17%	54%	0.5%	0.16%	0.09%	0.28%				
Mar-16	32%	36%	33%	1.3%	0.43%	0.48%	0.44%				
Apr-16	36%	21%	7%	1.3%	0.46%	0.27%	0.09%				
May-16	38%	10%	0%	3.6%	1.37%	0.36%	0.00%				
Jun-16	42%	21%	8%	2.0%	0.83%	0.41%	0.16%				
Jul-16	35%	36%	57%	4.2%	1.45%	1.49%	2.37%	10.76%	4.34%	2.86%	3.27%
Aug-16	35%	64%	41%								

As of: August 3, 2016

Evolution of IO Beta Allocation RV Variables

Date	MthlyHedgedReturn	OAS	Carry	RealizedVol	YCSlope	PrepayRiskPrem	PxCpnMult	LIB1m	RefiPctPool	RefiPctLoan	RefiIndex	CC	CCChange
1/30/2015	-8.94	65	1.10	60	121	103	4.78	0.17	0.66	0.66	2585	2.45	-38
2/27/2015	5.53	-87	1.62	73	138	89	5.51	0.17	0.61	0.56	1740	2.74	28
3/31/2015	1.18	-97	0.99	78	138	92	5.37	0.18	0.64	0.62	1942	2.66	-8
4/30/2015	1.19	28	0.24	69	147	81	5.56	0.18	0.64	0.61	1559	2.75	9
5/29/2015	-1.66	27	0.37	68	149	73	5.66	0.18	0.55	0.50	1234	2.81	6
6/30/2015	-0.66	10	0.32	75	170	51	6.16	0.19	0.45	0.40	1246	3.07	26
7/31/2015	1.18	14	0.19	82	153	66	5.93	0.19	0.50	0.42	1418	2.92	-16
8/31/2015	-1.53	21	0.49	76	146	68	5.92	0.20	0.55	0.53	1546	2.93	2
9/30/2015	-1.59	40	0.61	67	142	71	5.70	0.19	0.52	0.49	1590	2.80	-13
10/30/2015	-0.13	42	0.56	66	142	68	5.69	0.19	0.58	0.55	1535	2.84	4
11/30/2015	1.78	3	0.55	58	129	57	5.89	0.24	0.48	0.45	1489	2.95	10
12/31/2015	1.36	5	0.59	61	122	53	5.97	0.43	0.45	0.41	1667	3.01	6
1/29/2016	-2.11	21	0.74	57	116	74	5.38	0.43	0.62	0.60	1734	2.65	-36
2/29/2016	0.52	-7	1.58	67	95	92	5.07	0.44	0.67	0.63	2100	2.53	-12
3/31/2016	1.33	-6	1.43	65	105	93	5.16	0.44	0.67	0.63	1672	2.54	1
4/29/2016	1.29	-10	1.00	63	105	94	5.20	0.44	0.72	0.64	1910	2.56	2
5/31/2016	3.6	-49	0.99	56	96	91	5.23	0.47	0.69	0.65	1770	2.59	3
6/30/2016	1.97	-21	1.46	59	90	115	4.65	0.47	0.84	0.80	2081	2.25	-34
7/29/2016	4.15	28	1.56	59	79	112	4.44	0.50	0.82	0.75	2333	2.27	2